Northbrick3 application:
Categories exercise lab

Northbrick3 application

```
public class ProductData {
    String      productId;
    String productName;
    int     supplierId;
    String companyName;
    float     unitPrice;

    ProductData (String productId, String productName, int suppli
        this.productId   = productId;
        this.productName  = productName;
        this.supplierId   = supplierId;
        this.companyName = companyName;
        this.unitPrice = unitPrice;
    }
    public static Vector<ProductData> getProductList(Connection c
        Vector<ProductData> vec = new Vector<ProductData>();
        String sql = "Select ProductId, ProductName, Products.Sup
        sql += " WHERE Products.SupplierId = Suppliers.SupplierId
        System.out.println("getProductList: " + sql);
        try {
            Statement statement=connection.createStatement();
            ResultSet result = statement.executeQuery(sql);
            while(result.next()) {
```

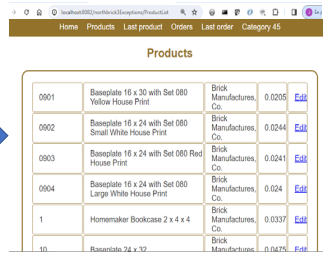
ResultSet object

	Col - 1	Col - 2	Col - 3
Row-1			
Row-2			
Row-3			

ResultSet pointer

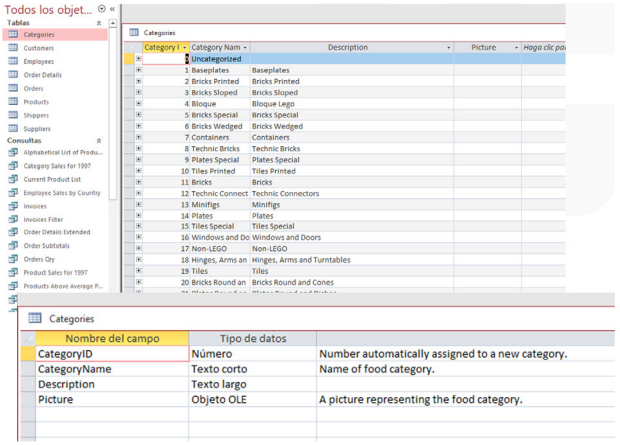
https://blog.csdn.net/allway2

Java Class
for managing the information
(MODEL)

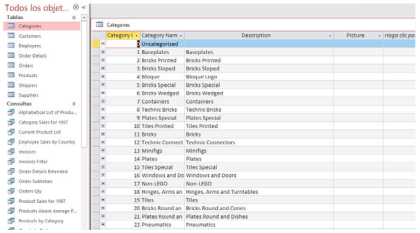


Servlet
for visualizing the information
(VIEW + CONTROLLER

Category List exercise



Category List exercise



ResultSet object

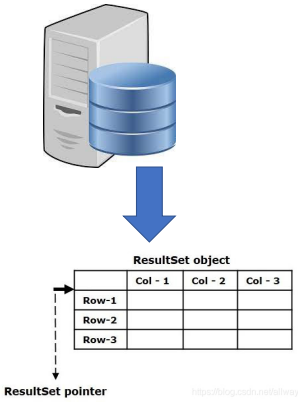
	Col - 1	Col - 2	Col - 3
Row-1			
Row-2			
Row-3			

ResultSet pointer

https://blog.csdn.net/allway2

Select * from Categories;

Category List exercise



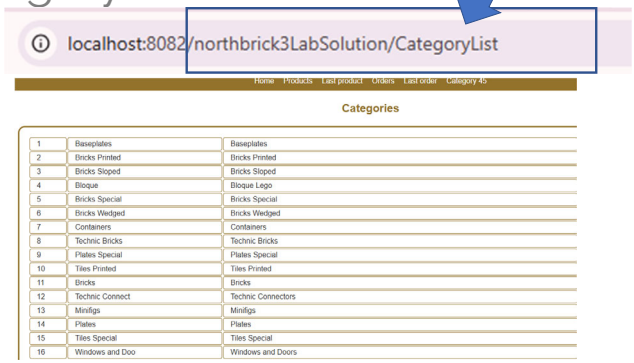

```
public class CategoryData {
    int    categoryId;
    String categoryName;
    String    description;

    CategoryData (int categoryId, String categoryName, String description) {
        this.categoryId    = categoryId;
        this.categoryName  = categoryName;
        this.description = description;
    }
    public static Vector<CategoryData> getCategoryList(Connection connection) {
        Vector<CategoryData> vec = new Vector<CategoryData>();
        String sql = "Select CategoryId, CategoryName, Description FROM Categories";
        System.out.println("getCategoryList: " + sql);
        try {
            Statement statement=connection.createStatement();
            ResultSet result = statement.executeQuery(sql);
            while(result.next()) {
                CategoryData category = new CategoryData(
                    Integer.parseInt(result.getString("CategoryId")),
                    result.getString("CategoryName"),
                    result.getString("Description")
                );
                vec.addElement(category);
```

Java Class for managing the information (MODEL)

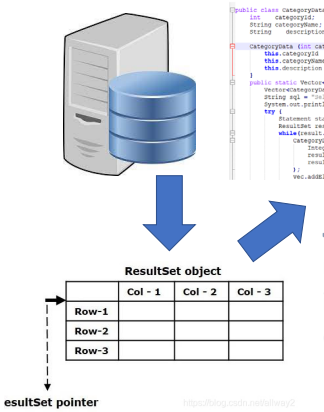
Category List exercise

Servlet name



Java Servlet to show the information (VIEW + CONTROLLER)

Category List exercise



Java Class for managing the information (MODEL)

Servlet for visualizing the information (VIEW + CONTROLLER

Add a link to categories table

To see the products linked to that category

Home Products Last product Orders Last order Category 45

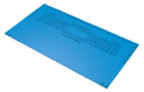
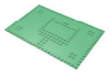
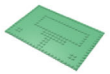
Categories

1	Baseplates	Baseplates	Products
2	Bricks Printed	Bricks Printed	Products
3	Bricks Sloped	Bricks Sloped	Products
4	Bloque	Bloque Lego	Products
5	Bricks Special	Bricks Special	Products
6	Bricks Wedged	Bricks Wedged	Products
7	Containers	Containers	Products
8	Technic Bricks	Technic Bricks	Products
9	Plates Special	Plates Special	Products
10	Tiles Printed	Tiles Printed	Products
11	Bricks	Bricks	Products

localhost:8082/northbrick3LabSolution/ProductList?id=1

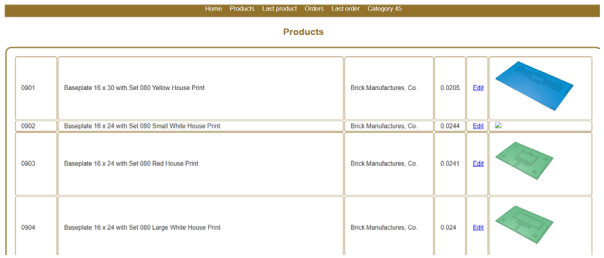
Home Products Last product Orders Last order Category 45

Products

0901	Baseplate 16 x 30 with Set 080 Yellow House Print	Brick Manufactures, Co.	0.0205	Edit	
0902	Baseplate 16 x 24 with Set 080 Small White House Print	Brick Manufactures, Co.	0.0244	Edit	
0903	Baseplate 16 x 24 with Set 080 Red House Print	Brick Manufactures, Co.	0.0241	Edit	
0904	Baseplate 16 x 24 with Set 080 Large White House Print	Brick Manufactures, Co.	0.024	Edit	

localhost:8082/northbrick3LabSolution/ProductList?id=1

```
String categoryId = req.getParameter("id");
```



Add a link to edit categories

Home Products Last product Orders Last order Category 45

Categories

1	Baseplates	Baseplates	Products	Edit
2	Bricks Printed	Bricks Printed	Products	Edit
3	Bricks Sloped	Bricks Sloped	Products	Edit
4	Bloque	Bloque Lego	Products	Edit
5	Bricks Special	Bricks Special	Products	Edit
6	Bricks Wedged	Bricks Wedged	Products	Edit
7	Containers	Containers	Products	Edit
8	Technic Bricks	Technic Bricks	Products	Edit
9	Plates Special	Plates Special	Products	Edit
10	Tiles Printed	Tiles Printed	Products	Edit

localhost:8082/northbrick3LabSolution/CategoryEdit?id=5

Home Products Last product Orders Last order Category 45

Category Form

Id	5
Name	Bricks Special
Description	Bricks Special

Enviar

```
public void doGet(HttpServletRequest req, HttpServletResponse res) throws ServletException, IOException {
    String idStr = req.getParameter("id");
    CategoryData category = CategoryData.getCategory(connection, idStr);
    toClient.println("<form action='CategoryUpdate' method='GET'>");
    toClient.println("<input name='categoryId' type='hidden' value='" + category.categoryId + "'></input>");
    toClient.println("<table border='1'>");
    toClient.println("<tr><td>Id</td>");
    toClient.println("<td>" + category.categoryId + "</td></tr>");
    toClient.println("<tr><td>Name</td>");
    String name = category.categoryName;
    toClient.println("<td><input name='categoryName' value='" + name + "'></td></tr>");
    toClient.println("<tr><td>Description</td>");
    toClient.println("<td><input name='description' value='" + category.description + "'></td></tr>");
    toClient.println("</table>");
    toClient.println("<input type='submit'>");
    toClient.println("</form>");
}
```

Read the category from url

Read the information for this category from the database

Form to send the info to the Category Update method

Home Products Last product Orders Last order Category 45

Category Form

Id	5
Name	Bricks Special
Description	Es un ejemplo

Enviar

When I click on Enviar, CategoryUpdate will be executed (method we defined in the form action)

localhost:8082/northbrick3LabSolution/CategoryUpdate?categoryId=5&categoryName=Bricks+Special&description=Es+un+ejemplo

http://localhost:8082/northbrick3LabSolution/CategoryUpdate?categoryId=5&categoryName=Bricks+Special&description=Es+un+ejemplo

Categories updated

1	Baseplates	Baseplates	Products	Edit
2	Bricks Printed	Bricks Printed	Products	Edit
3	Bricks Sloped	Bricks Sloped	Products	Edit
4	Bloque	Bloque Lego	Products	Edit
5	Bricks Special	Es un ejemplo	Products	Edit
6	Bricks Wedged	Bricks Wedged	Products	Edit
7	Containers	Containers	Products	Edit
8	Technic Bricks	Technic Bricks	Products	Edit
9	Plates Special	Plates Special	Products	Edit
10	Tiles Printed	Tiles Printed	Products	Edit
11	Bricks	Bricks	Products	Edit
12	Technic Connect	Technic Connectors	Products	Edit
13	Minifigs	Minifigs	Products	Edit

http://localhost:8082/northbrick3LabSolution/CategoryUpdate?categoryId=5&categoryName=Bricks+Special&description=Es+un+ejemplo

```
public class CategoryUpdate extends HttpServlet {

  public void doGet(HttpServletRequest req, HttpServletResponse res) throws ServletException  {

    String idStr = req.getParameter("categoryId");

    CategoryData category = new CategoryData(Integer.parseInt(idStr),
            req.getParameter("categoryName"),
            req.getParameter("description")
        );

    int n = CategoryData.updateCategory(connection, category);

}
```

Reads the categoryId from url

Creates a variable/object from CategoryData

Calls the method that is going to update the database